

MATH-1014 T10A

Tutorial 2

WANG Yixiao

Department of Mathematics

February 17, 2025



Table of contents

- 1 Volume of Revolution
- 2 Average Value
- 3 Arc Length & Symmetry

Review: Cylindrical Shell Method

Formula (Rotation about vertical axis)

$$V = \int_a^b \underbrace{2\pi r(x)}_{\text{Circumference}} \cdot \underbrace{h(x)}_{\text{Height}} \cdot \underbrace{dx}_{\text{Thickness}}$$

Strategy for Shells:

- **Cut Direction:** Parallel to rotation axis.
- **Radius $r(x)$:** Distance from the axis (x , $x - k$, etc.).
- **Height $h(x)$:** $y_{\text{top}} - y_{\text{bottom}}$.

Problem 1: Shell Method

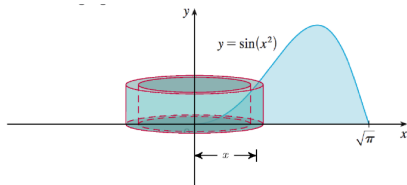
Problem 1. Let S be the solid obtained by rotating the region under $y = \sin(x^2)$ on $[0, \sqrt{\pi}]$ about the y -axis.

Key Info

- **Axis:** y -axis
- **Method:** Shells
- **Range:** $[0, \sqrt{\pi}]$

Hint:

Identify radius $r(x)$ and height $h(x)$.



Solution 1

Setup

Since we rotate about the y -axis, use shells with thickness dx :

$$r(x) = x, \quad h(x) = \sin(x^2)$$

$$V = \int_0^{\sqrt{\pi}} 2\pi x \sin(x^2) dx$$

Calculation: Let $u = x^2$, then $du = 2x dx$.

- Limits: $x = 0 \rightarrow u = 0$; $x = \sqrt{\pi} \rightarrow u = \pi$.

$$\begin{aligned} V &= \pi \int_0^{\pi} \sin u \, du \\ &= \pi \left[-\cos u \right]_0^{\pi} \\ &= \pi \left(-\cos \pi - (-\cos 0) \right) \\ &= \pi(1 + 1) = \mathbf{2\pi} \end{aligned}$$

Review: Disk Method

Formula (Hollow Rotation)

$$V = \pi \int_a^b ([R_{\text{outer}}]^2 - [r_{\text{inner}}]^2) (\text{thickness})$$

Visual Check:

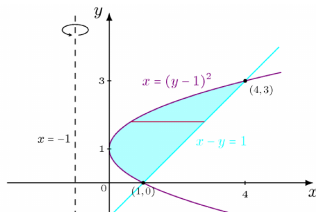
- **Cut Direction:** Perpendicular to the rotation axis.
- **Radii:** Always measure from the **axis of rotation**.

Problem 2: Disk Method

Problem 2. Region bounded by $x = (y - 1)^2$ and $x - y = 1$ is rotated about $x = -1$.

Analyze:

- ❶ Intersections?
- ❷ Inner/Outer radii relative to $x = -1$?
- ❸ Integrate w.r.t y or x ?



Note: The axis is offset ($x = -1$), not the y-axis.

Solution 2

Step 1: Intersections $(y - 1)^2 = y + 1 \implies y^2 - 3y = 0 \implies y = 0, 3.$

Step 2: Radii (Distance from $x = -1$)

- $R_{\text{outer}} = (y + 1) - (-1) = y + 2$ (Line)
- $r_{\text{inner}} = (y - 1)^2 - (-1) = (y - 1)^2 + 1$ (Parabola)

Step 3: Integral

$$\begin{aligned}
 V &= \pi \int_0^3 \left((y + 2)^2 - [(y - 1)^2 + 1]^2 \right) dy \\
 &= \pi \int_0^3 (12y - 7y^2 + 4y^3 - y^4) dy \\
 &= \pi \left[6y^2 - \frac{7}{3}y^3 + y^4 - \frac{1}{5}y^5 \right]_0^3 = \frac{117\pi}{5}
 \end{aligned}$$

Problem 3: Average Value

Recall: Average Value Formula

$$f_{\text{ave}} = \frac{1}{b-a} \int_a^b f(x) dx$$

Problem. Find the average value of

$$g(t) = \frac{t}{\sqrt{3+t^2}} \quad \text{on} \quad [1, 3].$$

Strategy:

- Don't forget the factor $\frac{1}{b-a}$.
- Observe that the numerator is related to the derivative of the inner function.

Solution 3

$$g_{\text{ave}} = \frac{1}{3-1} \int_1^3 \frac{t}{\sqrt{3+t^2}} dt = \frac{1}{2} \int_1^3 t(3+t^2)^{-1/2} dt$$

Substitution: Let $u = 3 + t^2$, then $du = 2t dt \implies t dt = \frac{1}{2} du$.

- $t = 1 \rightarrow u = 4$

- $t = 3 \rightarrow u = 12$

Calculation:

$$\begin{aligned} \text{Integral} &= \frac{1}{2} \int_4^{12} u^{-1/2} \left(\frac{1}{2} du \right) \\ &= \frac{1}{4} \left[2u^{1/2} \right]_4^{12} \\ &= \frac{1}{2} (\sqrt{12} - \sqrt{4}) = \frac{1}{2} (2\sqrt{3} - 2) = \sqrt{3} - 1 \end{aligned}$$

Review: Arc Length

Arc Length Formula

If $y = f(x)$ on $[a, b]$, then

$$L = \int_a^b \sqrt{1 + (f'(x))^2} dx.$$

Workflow:

- Compute $f'(x)$.
- Simplify $\sqrt{1 + (f')^2}$ using identities (e.g. $1 + \tan^2 x = \sec^2 x$).

Problem 4: Arc Length

Problem. Find the exact length of the curve:

$$y = \ln(\sec x), \quad 0 \leq x \leq \frac{\pi}{4}.$$

Formula

$$L = \int_a^b \sqrt{1 + (f'(x))^2} dx$$

Recall Trig Identity: $1 + \tan^2 x = \sec^2 x$.

Solution 4

1. Derivative:

$$y' = \frac{d}{dx} \ln(\sec x) = \frac{\sec x \tan x}{\sec x} = \tan x$$

2. Integrand:

$$\sqrt{1 + (y')^2} = \sqrt{1 + \tan^2 x} = \sqrt{\sec^2 x} = \sec x$$

(Note: $\sec x > 0$ for $0 \leq x \leq \pi/4$)

3. Integration:

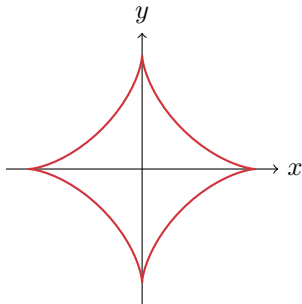
$$\begin{aligned} L &= \int_0^{\pi/4} \sec x \, dx \\ &= \left[\ln |\sec x + \tan x| \right]_0^{\pi/4} \\ &= \ln(\sqrt{2} + 1) - \ln(1 + 0) \\ &= \ln(\sqrt{2} + 1) \end{aligned}$$

Problem 5: Symmetry

Problem. Sketch $x^{2/3} + y^{2/3} = 1$ and find its total length.

Symmetry Argument:

- Equation is even in both x and y .
- Symmetric about both axes.
- **Total Length**
 $= 4 \times (\text{1st Quadrant Length}).$



Solution 5

1. Setup (1st Quadrant): $y = (1 - x^{2/3})^{3/2}$ for $x \in [0, 1]$.

$$\frac{dy}{dx} = \frac{3}{2}(1 - x^{2/3})^{1/2} \cdot \left(-\frac{2}{3}x^{-1/3}\right) = -x^{-1/3}(1 - x^{2/3})^{1/2}$$

2. Simplify:

$$1 + (y')^2 = 1 + x^{-2/3}(1 - x^{2/3}) = 1 + x^{-2/3} - 1 = x^{-2/3}$$

So, $\sqrt{1 + (y')^2} = x^{-1/3}$.

3. Integrate:

$$L_1 = \int_0^1 x^{-1/3} dx = \left[\frac{3}{2}x^{2/3}\right]_0^1 = \frac{3}{2}$$

$$\text{Total Length } L = 4 \times L_1 = 4 \times \frac{3}{2} = 6$$

Problem 6: Parametric Arc Length

Problem. Find the exact arc length of the parametric curve

$$x(t) = e^t \cos t, \quad y(t) = e^t \sin t, \quad 0 \leq t \leq \frac{\pi}{2}.$$

Formula (Parametric)

$$L = \int_{\alpha}^{\beta} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

Hint

Compute $\frac{dx}{dt}$, $\frac{dy}{dt}$ and try to simplify $\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2$.

Solution 6

1. Derivatives:

$$\frac{dx}{dt} = e^t(\cos t - \sin t), \quad \frac{dy}{dt} = e^t(\sin t + \cos t).$$

2. Speed:

$$\begin{aligned} \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 &= e^{2t}[(\cos t - \sin t)^2 + (\sin t + \cos t)^2] \\ &= e^{2t}[(\cos^2 t - 2\sin t \cos t + \sin^2 t) + (\sin^2 t + 2\sin t \cos t + \cos^2 t)] \\ &= e^{2t} \cdot 2(\sin^2 t + \cos^2 t) = 2e^{2t}. \end{aligned}$$

Hence,

$$\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = \sqrt{2}e^t.$$

3. Arc length:

$$L = \int_0^{\pi/2} \sqrt{2}e^t dt = \sqrt{2}[e^t]_0^{\pi/2} = \sqrt{2}(e^{\pi/2} - 1).$$

Extension

Problem. Sketch $x^{2/3} + y^{2/3} = 1$ and find its total length.

How to solve this problem using parametric formula?

THANK YOU!

